Collaborative Design of a Silicon Photonic Integrated Circuit: Laser-Driven Optical Resonators with Fabry-Perot Cavities

Aryan Chahardovalee
Department of Electrical and Computer Engineering
University of British Columbia

1 Intoduction

In this project, we aim to create a single photonic integrated circuit (PIC) that incorporates each student's optimal optical resonator, designed in Project 1, with a commercially available off-the-shelf laser powering all resonators. As part of the ELEC 413 course on Semiconductor Lasers at the University of British Columbia (UBC), students have the opportunity to engage with the UBC-led SiEPICfab consortium, which provides access to cutting-edge silicon photonics fabrication processes.

The design approach for this project involves creating a Fabry-Perot cavity resonator by sandwiching a waveguide between two equally-lengthed Bragg gratings. The high-quality factor of the cavity is ensured through simulations and calculations performed in Project 1, and as compared to Project 1, the cavity length has been sufficiently increased to allow for a reduced waveguide loss. For more information, please refer to the design document of Project 1.

Another essential consideration for this project is the fabrication process bias, wherein waveguides with a specified width of 350 nm will be printed at 315 nm, 35 nm smaller than the intended width. To account for this biasing effect, the Fabry-Perot cavity in this project is designed for a width of 335 nm and drawn for fabrication at 370 nm. After performing the analysis on the Fabry-Perot Bragg cavity resonators from Project 1, it was discovered that design ARYANC1_3 had the highest quality factor of 21000. However, the resonator outlined in this project corresponds to design ARYANC1_1 which was initially assumed to have the highest quality factor among all of the Project 1 designs. This project is still a good attempt at integrating the laser and the PIC chip as design ARYANC1_1 still provides a sufficient quality factor for our frequency of operation.

Name of Parameter	**Value of Parameter**
Frequency of Operation	228.849 THz
Width	Actual: 335 nm / Fabricated: 370 nm
Corrugation Width	30 nm
Thickness	220 nm
Bragg Number	103
Waveguide Cavity Length (between Bragg Gratings)	1695 microns
Bragg Period	269 nm

Table 1: Geometrical Parameters Summary

2 Simulation

The simulations in this project consist of two parts. One where the laser is going to be analyzed and modeled and another to analyze the resonators effects on the laser output.

2.1 Laser

To analyze the laser we will use the data that was obtained as part of Lab 2. A MATLAB model of the laser with parameters such as quantum efficiency, modal gain (s(-1)), transparency carrier number, spontaneous emission rate (s(-1)), carrier lifetime (s) and photon lifetime (s) is used to model the light input from the commercial laser. This process is described in the Appendix (Appendix B - Lab 2) section of this document. The MATLAB circuit model is shown in blue in Figure 1, superimposed on the reflection data that was obtained from Design_AryanC in Lab 2.

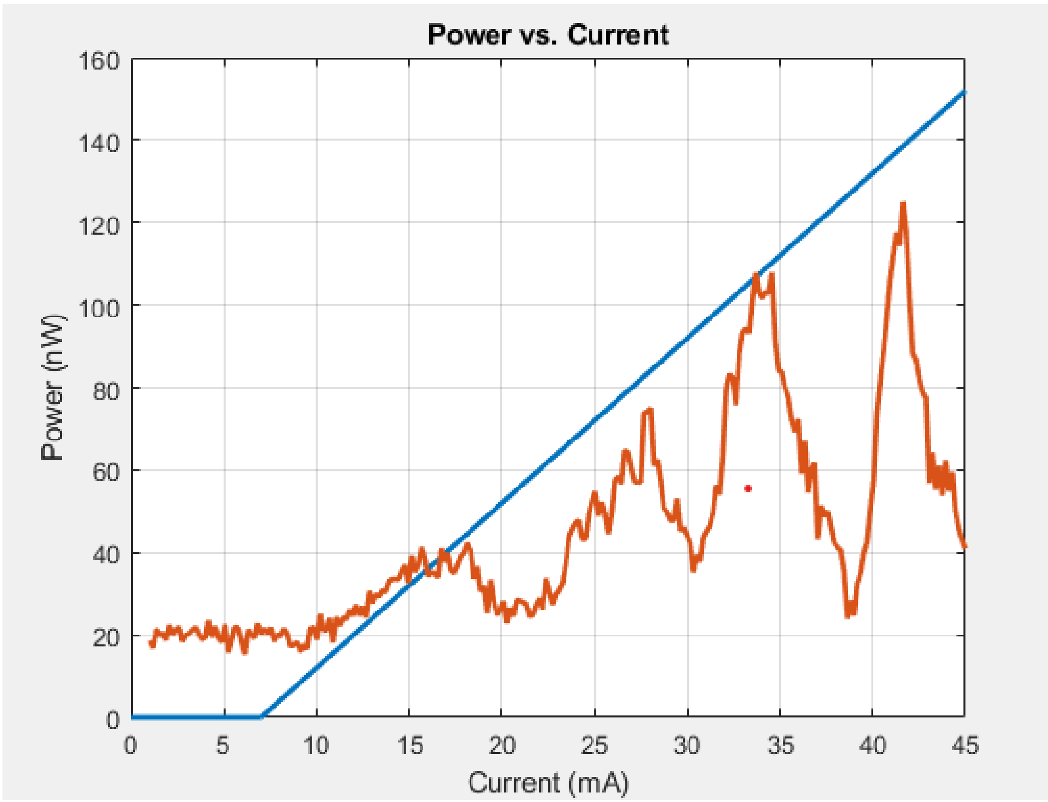


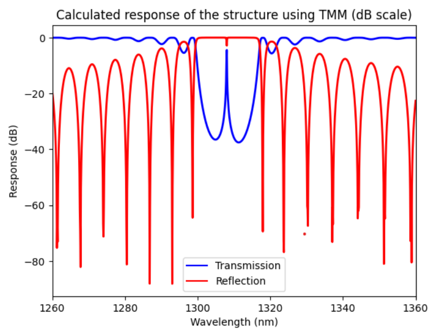
Figure 1: Laser Model for the Reflection Output Light - From Lab 2

2.2 Resonator

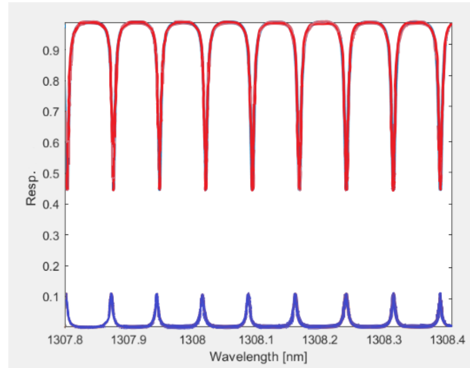
The Fabry-Perot Bragg cavity resonator is modelled as in Project 1. The characteristics of this resonator (Figure 2) are specified in Table 1. Figure 2.b specifies the brag transmission and reflection characteristics in the region where the resonator operates. Figure 2.b has a higher resolution than Figure 2.a and is represented in relative reflectivity and transmissivity values between 0 and 1. In Figure 2.a, the red lines specify reflection and the blue lines indicate the transmission spectrum.

2.3 Laser + Resonator

To combine the Laser model with the resonator model at the specified wavelength of operation, the two signals From Figure 3 are superimposed on the laser model (Figure 1 - Blue Line) by convoluting the two signals. The results that are obtained are the predicted transmission and reflection spectrum that is expected to be received from the optical circuit upon performing measurements in Lab 2. These spectra will be compared to the transmission and reflection spectrum in section 5 of this report. It should be mentioned that the simulations for Laser + Resonator are missing some crucial information that may possibly result in our experimental data deviating from the predicted behaviour. These deviations come from the fact that we did not account for the effects of reflection from the grating couplers and/or any other waveguides or optical devices in between the laser and the resonators.



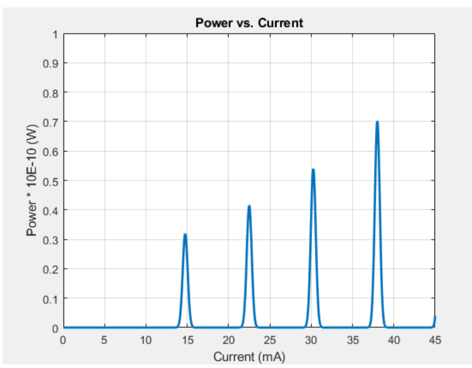
(a) Transmission and Reflection Spectrum of Fabry Perot Bragg Cavity Resonator with a Long Waveguide



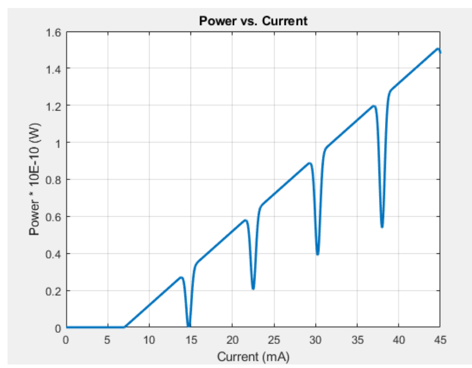
(b) Transmission and Reflection Spectrum of Fabry Perot Bragg Cavity Resonator with a Long Waveguide - Zoomed In

Figure 2: Transmission and Reflection Spectra for Resonator + Laser

Further investigation is required in order to be able to model the Laser + Resonator system more accurately.



(a) Transmission Spectrum of Fabry Perot Bragg Cavity Resonator + Laser in Watt scale



(b) Reflection Spectrum of Fabry Perot Bragg Cavity Resonator + Laser in Watt scale

Figure 3: Predicted Transmission and Reflection Spectra

3 Fabrication

The fabrication process used in this report is based on the SiEPICfab-EBeam-ZEP-PDK, which is specifically designed for the ZEP process. The device was fabricated using 100 keV Electron Beam Lithography [1] on an SOI (Silicon-On-Insulator) wafer, consisting of a 220 nm thick silicon layer. The wafer is 200 mm in diameter and uses a CZ-grown P-type Boron-doped top silicon layer with a resistivity of 30-60 ohm-cm. The silicon layer is formed on a BOX (Buried Oxide) layer that is 3500 nm thick, followed by a handle layer which is 725 µm thick. [1]

The baseline process involves a single full etch using a positive resist, ZEP, with no cladding. ZEP 520A, which is 550 nm thick, is used for process documentation, including process bias. For state-of-the-art component information, ZEP 520A-7 with a thickness of 300 nm is used. [1]

After a solvent rinse, nanostrip surface treatment, and hot-plate dehydration bake, ZEP 520A was spin-coated at 2100 rpm, then hotplate baked at 180 °C for 2 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 0.5nA beam current, and 500µm exposure field size. The machine grid used for shape placement was 1nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 2nm. An exposure dose of 230µC/cm2 was used. [1]

The resist was developed by immersion in the developer ZED-N50 for 1 minute, followed by a rinse in isopropanol 1 minute, and then blown dry with nitrogen. The silicon was removed from exposed areas using inductively coupled plasma etching in an Oxford Cobra, with a SF6 gas flow of 25 sccm and CF4 gas flow of 30sccm, pressure of 10mT, ICP power of 800W, and a platen temperature of 15°C, resulting in a bias voltage of 220V. During etching, chips were mounted on a 100mm silicon carrier wafer vacuum grease. Total etch time was 5 minutes. The unexposed resist was stripped using UV exposure followed by a dip into warm MICROPOSIT Remover 1165 and post-etch residue remover EKC 265 followed by IPA rinse and nitrogen dry. [1],[5],[6],[7]

Next, the chips were oxygen plasma treated and baked at 120°C as dehydration bake to spin coat the positive photoresist AZP4620 at 4500RPM. This was to pattern the trench layer using the maskless aligner (MLA150) at 365nm wavelength with an exposure dose of 1000mJ/cm2 and defocus of -4. The exposed resist was removed by dipping into AZ MIF 300 for 4 minutes followed by a 1-minute rinse in de-ionized water. The exposed trench pattern then underwent various etch recipes to remove the top 220nm silicon layer, 2um of buried oxide layer, and roughly 80um/90um of the bottom substrate silicon layer. The substrate silicon layer was etched using 15 loops of the Bosch cycle process to give a vertical sidewall. This step was required to match the laser die thickness (which was about 100um) to the device layer on the chip as we placed the laser into the trench/hole region. This whole deep etching process was carried out at the 4D labs in Simon Fraser University (SFU) using the deep reactive ion etcher, the Rapier™ DRIE system. [1],[5],[6],[7]

The exposed trench region was then metalized using the Ebeam evaporation tool, the Deedirector at UBC. A 10 nm layer of titanium as an adhesion layer followed by 200 nm of gold was deposited onto the substrate. The resist was then stripped using warm Remover 1165 (80°C) solution for 1 hour while constantly in motion using a stir bar, followed by an IPA rinse and nitrogen blow dry.[1],[5],[6],[7]

For the purpose of assembly, the chip was diced into 5mm squares. The target patterned chip was then surface treated with oxygen plasma. Tin-gold preforms were used to die bond the MACOM laser into the deep trench area, aligning the laser facet to the surface coupler on the chip. The Tresky die bonder tool was used to perform the pick-and-place and temperature control during the bonding. Lastly, the Vangaurd photonic wire bonder was used to connect the surface taper on the chip to the laser facet interface. A resist was first deposited at the target interface location. The tool used image analysis to detect the facets of the laser and the silicon coupling region, creating a model. The femtosecond laser in the tool then exposed the resist with a suitable recipe consisting of the speed and exposure along the created drawing. The unexposed resist was removed through the development process. With a final rinse in IPA, the chip was ready for testing. In the interest of time and experiment goals, electrical wire bonding on the laser was not performed this time, but needle probes were used for testing instead.[1],[5],[6],[7]

This fabrication process includes various layers for drawing silicon waveguides, etch trenches, metal routing, heaters, deep trench etch regions, and more. To ensure proper fabrication, designers draw EBL (Electron Beam Lithography) Write Field Regions in rectangles with a maximum size of 1x1 mm. Designers can choose their own fields to avoid field stitching boundaries or to determine where the boundaries are placed. A DRC (Design Rule Check) is performed to ensure the EBL regions are defined correctly. With this fabrication process, high-quality photonic integrated circuits can be manufactured, enabling advancements in a wide range of applications, such as telecommunications, sensing, and computing. [1],[5],[6],[7]

The most outstanding design from Project 1 has been selected for integration into an optical circuit, incorporating the top-performing Fabry-Perot Bragg cavity designs. The process can be described as follows:

The most outstanding design from Project 1 has been selected for integration into an optical circuit, incorporating the top-performing Fabry-Perot Bragg cavity designs. The process can be described as follows:

- A single on-chip laser powers all resonators within the circuit. A splitter tree is employed to evenly distribute the light from the laser across all resonators.
- Each student's design consists of a Python script defining the Bragg grating and waveguide specifications. Pins are connected using the "connect_cell" command. The connection sequence in the model is described below:

```
        ...
#opt1<---inst_wg1--->opt2 + opt2<---cell_y--->opt1 = opt1
    <---inst_y1--->opt1
#opt1<---inst_y1--->opt1 + pin1<---cell_taper--->pin2 =
    opt1<---inst_taper1--->pin2
#opt1<---inst_taper1--->pin2 + opt1<---cell_bragg--->opt2
    = opt1<---inst_bragg1--->opt2
#opt1<---inst_bragg1--->opt2 + opt2<---cell_bragg--->opt1
    = opt1<---inst_bragg2--->opt1
        ...
```

In the Python code, the input corresponds to the "opt1" pin of inst_wg1. The waveguide's transmission output corresponds to the "pin1" of inst_tapper2, while the reflection output is the "opt3" pin of inst_y1 (not shown above). Inst_y1 represents the input and the Y-branch section of the circuit. The Y-branch is utilized to relay reflectivity information back to the detectors, which are connected to the "opt3" pin of inst_y1. A taper is incorporated between the Y-branch and the first Bragg grating, enabling the width to transition from the standard 350 nm to 370 nm. This taper is instantiated twice for both sides of the Bragg gratings.

- Finally, a long waveguide is added to the design. This waveguide has a width of 370 nm and is located between the "opt2" pin of inst_bragg1 and the "opt2" pin of inst_bragg2 instance.

```
            ...
    try:
        connect_pins_with_waveguide(inst_bragg1, 'opt2',
            inst_bragg2, 'opt2', waveguide_type='Strip 1310
             nm, w=370 nm (core-clad)',
            turtle_A =
                [275,90,20,90,270,-90,20,-90,265,90,20,90,
            260,-90,20,-90, 255, 90, 20, 90, 250, -90, 20,
                 -90] )
            ...
```

- The full code for Design_AryanC is included in the Appendix section (Appendix A1). The connections that are required for integration of each design (ex. Design_AryanC, ...) into the main optical circuit are specified by the instructor and are specific to the fabrication process that is outlined above.

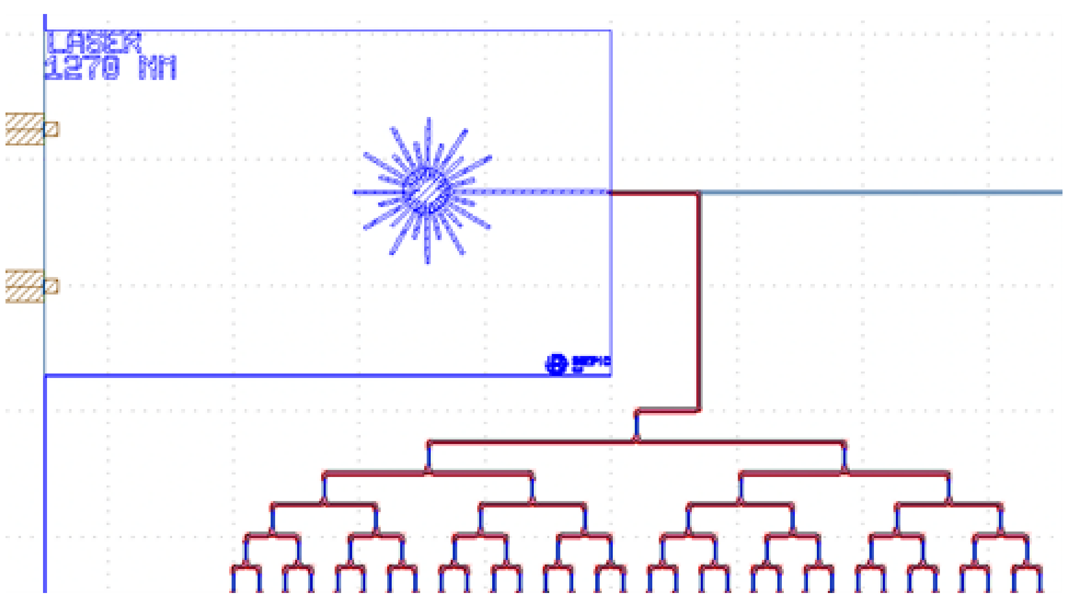
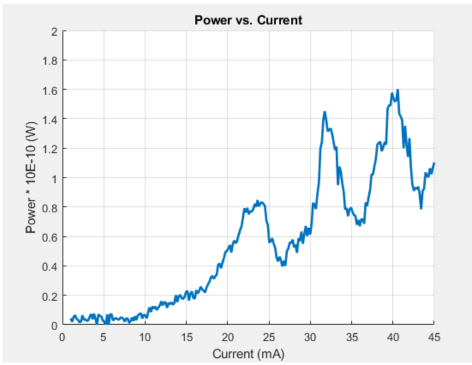


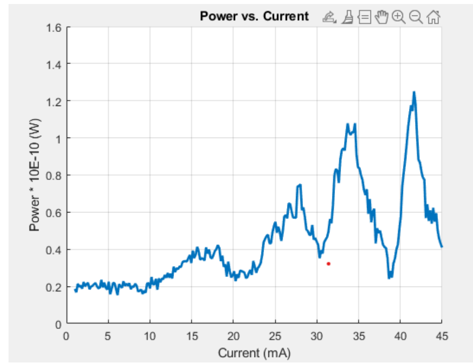
Figure 4: Input to ELEC 413 - Project 2 optical circuit

4 Test and Measurements

Upon receiving the fabricated circuit, a setup is developed by the instructor and the TA to facilitate the measurement process for each individual design. Students in groups of 5, measured and recorded the light output of several designs. Each pair of students were assigned 3 - 6 individual resonator designs and results were shared amongst students for further analysis. The summary of this process is included in Appendix B - Lab 2. Figure 5 depicts the transmission and reflection spectra that were obtained from measuring Design_AryanC's performance.



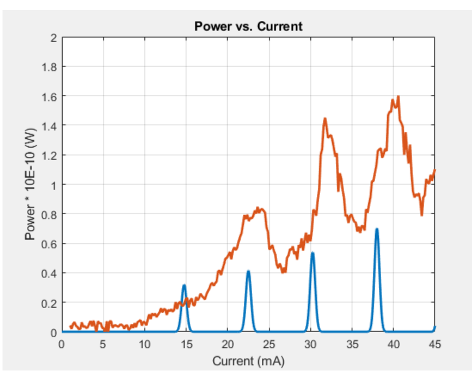
(a) Transmission Spectrum of Fabry Perot Bragg Cavity Resonator + Laser in Watt scale



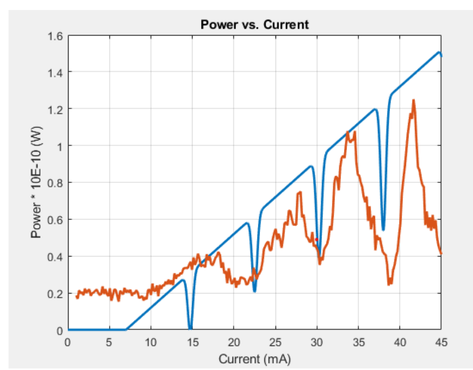
(b) Reflection Spectrum of Fabry Perot Bragg Cavity Resonator + Laser in Watt scale

Figure 5: Experimental Transmission and Reflection Spectra

5 Conclusion and Discussion



(a) Transmission Spectrum of Fabry Perot Bragg Cavity Resonator + Laser in Watt scale



(b) Reflection Spectrum of Fabry Perot Bragg Cavity Resonator + Laser in Watt scale

Figure 6: Comparing Experimental and Predicted Transmission and Reflection Spectra

When comparing the experimental L vs I graphs for both transmission and reflection with the predicted spectra using the laser + resonator model in MATLAB, we can observe that the resonance behaviour occurs at the same current interval for both the experimental and predicted plots. This is a promising indication that the model correctly predicts the behaviour of the entire system. However, it is important to note that the amplitude of the predicted model is often smaller than what was observed in the Lab 2 measurements. This discrepancy could be attributed to a number of factors, such as the current increasing and resulting in temperature increases, which may alter the wavelength and center frequency. Additionally, the setup intoduces some unknown effects. These unknown effects stem from the grating couplers and adiabatic splitters, which may cause additional reflections and contribute to the observed discrepancy. Another notable observation is that the pointy regions in the reflection spectrum appear flatter in the predicted model, while in the experimental design, they are more pointed. This could be due to a variety of factors, such as limitations in the modelling software, where grating couplers and splitters have been ignored. Furthermore, it is possible that the fabrication process itself may introduce variations that affect the experimental results. The quality of the etching, deposition of metals, and alignment of the laser and resonator may all contribute to differences between the predicted and experimental results. In summary, while the agreement between the experimental and predicted resonance behaviour is promising, it is important to acknowledge the limitations of the model and the potential for experimental variations to impact the results. Further investigation and refinement of the simulation model may be necessary to fully understand the observed discrepancies.

A 1

```
from pya import *

def design_AryanC(cell, cell_y, inst_wg1, inst_wg2, inst_wg3,
    waveguide_type):

    # load functions
    from SiEPIC.scripts import connect_pins_with_waveguide,
        connect_cell
    ly = cell.layout()
    library = ly.technology().name

    #####
    # designer circuit:

    # Create a physical text label so we can see under the
        microscope
    # How do we find out the PCell parameter variables?
    '''
    c = ly.create_cell('TEXT','Basic')
    [p.name for p in c.pcell_declaration().get_parameters() if c.
        is_pcell_variant]
    c.delete()
    '''
    # returns: ['text', 'font_name', 'layer', 'mag', 'inverse', '
        bias', 'cspacing', 'lspacing', 'eff_cw', 'eff_ch', '
        eff_lw', 'eff_dr', 'font']
    from SiEPIC.utils import get_technology_by_name
    TECHNOLOGY = get_technology_by_name(library)
    cell_text = ly.create_cell('TEXT', "Basic", {
        'text':cell.name,
        'layer': TECHNOLOGY['M1'],
        'mag': 30,
         })
    if not cell_text:
        raise Exception ('Cannot load text label cell; please
            check the script carefully.')
    cell.insert(CellInstArray(cell_text.cell_index(), Trans(Trans
        .R0, 25000,125000)))

    # load the cells from the PDK
    # choose appropriate parameters
    #####################TAPER##########################
    cell_taper = ly.create_cell('taper', library, {
        'wg_width1': 0.35,
        'wg_width2': 0.37,
            })
    if not cell_taper:
        raise Exception ('Cannot load taper cell; please check
            the script carefully.')
    #####################BRAGG##########################
    cell_bragg = ly.create_cell('Bragg_grating', library, {
        'number_of_periods':100,
        'grating_period': 0.269,
        'corrugation_width': 0.03,
        'wg_width': 0.370,
        'sinusoidal': False})
    if not cell_bragg:
        raise Exception ('Cannot load Bragg grating cell; please
            check the script carefully.')
    ###Y Branch and Input/Output waveguides to be imported###
    inst_y1 = connect_cell(inst_wg1, 'opt2', cell_y, 'opt2') #
        opt1<---inst_wg1--->opt2 + opt2<---cell_y--->opt1 = opt1
```

```
    <---inst_y1--->opt1 ????If Inst_y1 has 2 opt1 pins, then
    how do we know which one to connect to????????

inst_taper1 = connect_cell(inst_y1, 'opt1', cell_taper, 'pin1
    ') # opt1<---inst_y1--->opt1 + pin1<---cell_taper--->pin2
     = opt1<---inst_taper1--->pin2
inst_bragg1 = connect_cell(inst_taper1, 'pin2', cell_bragg, '
    opt1') # opt1<---inst_taper1--->pin2 + opt1<---cell_bragg
    --->opt2 = opt1<---inst_bragg1--->opt2
inst_bragg2 = connect_cell(inst_bragg1, 'opt2', cell_bragg, '
    opt2') # opt1<---inst_bragg1--->opt2 + opt2<---cell_bragg
    --->opt1 = opt1<---inst_bragg2--->opt1  ????If Bragg2 has
     2 opt1 pins, then how do we know which one to connect to
    ????????
inst_bragg2.transform(Trans(250000,120000))



#####
# Waveguides for the two outputs:
connect_pins_with_waveguide(inst_y1, 'opt3', inst_wg3, 'opt1
    ', waveguide_type=waveguide_type)

inst_taper2 = connect_cell(inst_bragg2, 'opt1', cell_taper, '
    pin2') # opt1<---inst_bragg2--->opt1 + pin2<---cell_taper
    --->pin1 = opt1<---inst_taper2--->pin1
connect_pins_with_waveguide(inst_taper2, 'opt1', inst_wg2, '
    opt1', waveguide_type=waveguide_type) # in case of
    further errors please change opt1 to pin1 :(

'''
make a long waveguide, back and forth,
target 0.2 nm FSR assuming ng = 4
> wavelength=1270e-9; ng=4; fsr=0.2e-9;
> L = wavelength**2/2/ng/fsr
> L * 1e6
> 1000 [microns]
using "turtle" routing
https://github.com/SiEPIC/SiEPIC-Tools/wiki/Scripted-Layout#
    adding-a-waveguide-between-components
'''
try:
     connect_pins_with_waveguide(inst_bragg1, 'opt2',
         inst_bragg2, 'opt2', waveguide_type='Strip 1310 nm, w
         =370 nm (core-clad)',
          turtle_A =
              [275,90,20,90,270,-90,20,-90,265,90,20,90,260,-90,

          20,-90, 255, 90, 20, 90, 250, -90, 20, -90] )
except:
     connect_pins_with_waveguide(inst_bragg1, 'opt2',
         inst_bragg2, 'opt2', waveguide_type='Strip 1310 nm, w
         =370 nm (core-clad)',
          turtle_A =
              [275,90,20,90,270,-90,20,-90,265,90,20,90,260,-90,

          20,-90, 255, 90, 20, 90, 250, -90, 20, -90] )

return inst_wg1, inst_wg2, inst_wg3
```

B Lab 2

B.1 Introduction and Purpose

The following procedure describes how the commercial laser that is used for Project 2 is modelled based on the L vs I output of the laser for both the transmission and reflection. The laser light is being measured after going through the optical circuit that is designed as part of project 2 and then is received by the HP8163A model and sent to a PC for further analysis. Measuring the light output of the laser after going through the optical circuit gives us a good understanding of the characteristics of the light that enters the circuit.

B.2 Setup

A Python script running on a PC will perform the measurements by changing the laser current input and consequently shining light on the ZEP-manufactured chip. The output light from both the transmission and the reflection outputs of the optical circuit is measured and sent back to the PC. Connections between measurement devices and the PC are accomplished by the use of VISA drivers. To control the current of the laser, we use an LDC501 (TEC + Laser Driver) provided by Stanford Research Systems. This device allows for precise control over the operating conditions of the laser, which is crucial for studying its characteristics under various conditions. The current is swept from 0 to 45 mA.

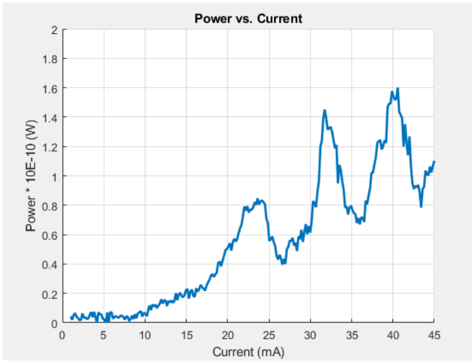
Figure 7: LDC501 (TEC + Laser Driver)

The optical circuit is placed on a platform which is calibrated by the instructor and TA to align the output light of the laser with the grating couplers that are fabricated on the chip. To measure each individual design, the position of the laser light has to be changed. This is achieved by controlling the setup using a PC Application specific to the setup.

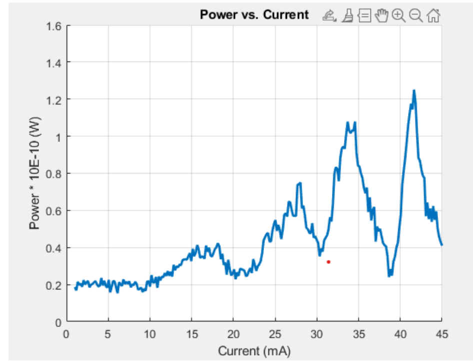
Finally, an HP8163A device is used to record the measurements and send them back to the Python script, running on the PC. The HP8163A Lightwave Multimeter is a high-performance, versatile instrument designed to measure optical power and perform loss measurements in fiber optic networks. The HP8163A can be connected to tunable laser sources to perform wavelength-dependent loss measurements or swept-wavelength measurements in wavelength-division multiplexing (WDM) systems. It also offers a user-friendly interface with an intuitive menu structure, making it easy to configure and operate the instrument.

Figure 8: HP8163A Lightwave Multimeter

B.3 Data and Analysis



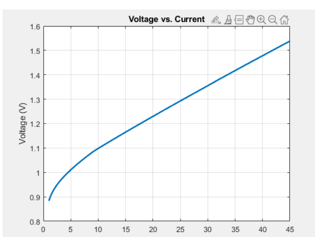
(a) Transmission Spectrum of Fabry Perot Bragg Cavity Resonator + Laser in Watt scale



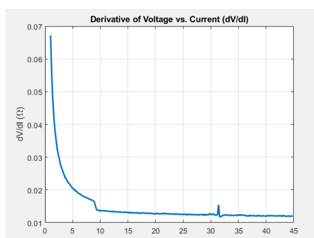
(b) Reflection Spectrum of Fabry Perot Bragg Cavity Resonator + Laser in Watt scale

Figure 9: Experimental Transmission and Reflection Spectra

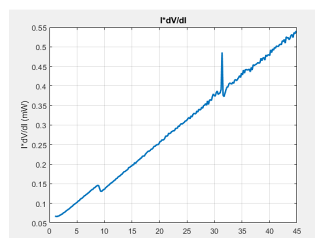
Figure 9 depicts the L vs I output for both transmission and reflection outputs of the optical circuit, Design_AryanC. The power is in the range of nWs. This is expected as the light input from the laser is split into multiple fragments using adiabatic Y splitters to be distributed evenly between all of the designs within the chip. Moreover, many oscillations are observed in the active region of the laser where light output is to increase linearly with increasing current. These oscillations can be the result of many factors, one of which is the resonator that we have incorporated into our optical circuit. Discussions on whether this data implies the effects of the resonator or not are included in Section 5 of the Project 2 document.



(a) V vs I Characteristics



(b) dV vs dI Characteristics



(c) dV vs dI Characteristics

Figure 10: Voltage vs Current Characteristics of the Laser

Voltage vs current characteristics of the laser is measured and shown above. The plot of dV/dI vs current and I*dV/dI is obtained by manipulating the data in MATLAB. The differential resistance is measured using Figure 10.b and the laser threshold current is found from Figure 10.c where the first transition is observed. The I*dV/dI graph provides a useful way to identify the threshold current of the laser because it represents the product of the current and the differential resistance (dV/dI). At the threshold current, the laser transitions from spontaneous emission (below threshold) to stimulated emission (above threshold), and there is a significant change in the device behaviour.

Below the threshold current, the differential resistance is relatively high, primarily due to the resistive properties of the device. The laser behaves more like a simple diode with no significant lasing action occurring. As a result, the I*dV/dI curve remains relatively flat below the threshold current, but it will be increasing as the differential resistance continues to decrease as we approach the threshold current.

When the current reaches the threshold current, the laser begins to emit light through stimulated emission. This process increases the photon density and, consequently, the carrier recombination rate within the laser cavity. The increase in the recombination rate leads to a sudden decrease in the differential resistance. The decrease in differential resistance causes the I*dV/dI curve to increase rapidly at the threshold current.

The I*dV/dI plot highlights this transition by showing a clear transition in the slope of the curve at the threshold current. This change corresponds to the point where the laser transitions from spontaneous to stimulated emission, making it easier to identify the threshold current of the laser.

Table 2 summarized the results obtained from these measurements:

Name of Parameter	Value of Parameter
Differential resistance at threshold	0.01 Ohm
Differential resistance at 2 x threshold	0.01 Ohm
Threshold Current	7.5 mA

Table 2: Lab 2 Results Summary

B.4 Discussion

In this lab report, we have investigated the properties of a commercial laser and its interactions with an optical circuit designed for Project 2. We have analyzed the voltage vs. current characteristics of the laser and identified its threshold current, as well as the differential resistance at the threshold and at 2x the threshold current. Additionally, we have studied the transmission and reflection spectra of the Fabry Perot Bragg Cavity Resonator and the laser.

The oscillations observed in the active region of the laser could be attributed to several factors. One possible explanation is the presence of the resonator in the optical circuit. The resonator may introduce multiple reflections and constructive/destructive interference, leading to the oscillatory behaviour observed in the light output. Other factors that may contribute to these oscillations include temperature variations, fluctuations in the laser's driving current, and imperfections in the fabrication of the optical circuit or its components.

The determination of the threshold current and differential resistance measurements offers valuable insights into the behaviour of lasers. These metrics help in identifying the point at which the laser transits from spontaneous emission to stimulated emission, thereby aiding in the modelling of the laser operation, which is crucial for Project 2. Figure 11 illustrates the reflection spectrum of Design_AryanC, where the MATLAB laser model for simulating the laser using rate equations is depicted by the blue line. The desired outcome of this simulation is to achieve the previously calculated laser threshold of 7.5 mA while modifying other parameters such as quantum efficiency, modal gain (s(-1)), transparency carrier number, spontaneous emission rate (s(-1)), carrier lifetime (s), and photon lifetime (s). Lower differential resistance values at the threshold and twice the threshold current signify enhanced efficiency in converting electrical energy into optical energy, which is highly desirable for various applications.

Figure 11: Laser Model for the Reflection Output Light

References

[1] Chrostowski, L., Hochberg, M. (2015). *Silicon Photonics Design: From Devices to Systems* (Chapter 12). Cambridge University Press.

[2] ELEC 413. (2023). Week 4: Bragg Gratings.

[3] ELEC 413. (2023). Lab 1.

[4] ELEC 413. (2023). Project 1.

[5] Pang, R. W. C., Baiocco, C., Chen, M. Y., Yang, T., Chrostowski, L., McMullin, J. N. (2018). A 1x8 wavelength-selective switch fabricated with a single patterned etch step in a silicon-on-insulator platform. Journal of Lightwave Technology, 36(23), 5638-5645. doi:10.1109/JLT.2018.2871021

[6] SiEPIC. (n.d.). SiEPICfab_Shuksan_PDK. GitHub. https://github.com/SiEPIC/SiEPICfab_Shuksan_PDK

[7] SiEPIC. (n.d.). SiEPICfab_EBeam_ZEP_PDK. GitHub. https://github.com/lukascubc/SiEPIC_EBeam_PDK/wiki/Installation-instructions

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